03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 17

(JCD 12.2)

Annelies Mortier - office PW G35 - a.mortier@bham.ac.uk

UNIVERSITY OF BIRMINGHAM

Previous lecture

- Stars leave the main sequence when H core burning stops
- Inert He cores are isothermal
- H shell burning makes the core contract and the star expand
- When a star reaches the Hayashi track again, it must increase in luminosity: red giant branch
- For a moderate-mass star, the core mass fraction exceeds the Schönberg-Chandrasekhar limit after which the star quickly evolves across the Hertzsprung gap

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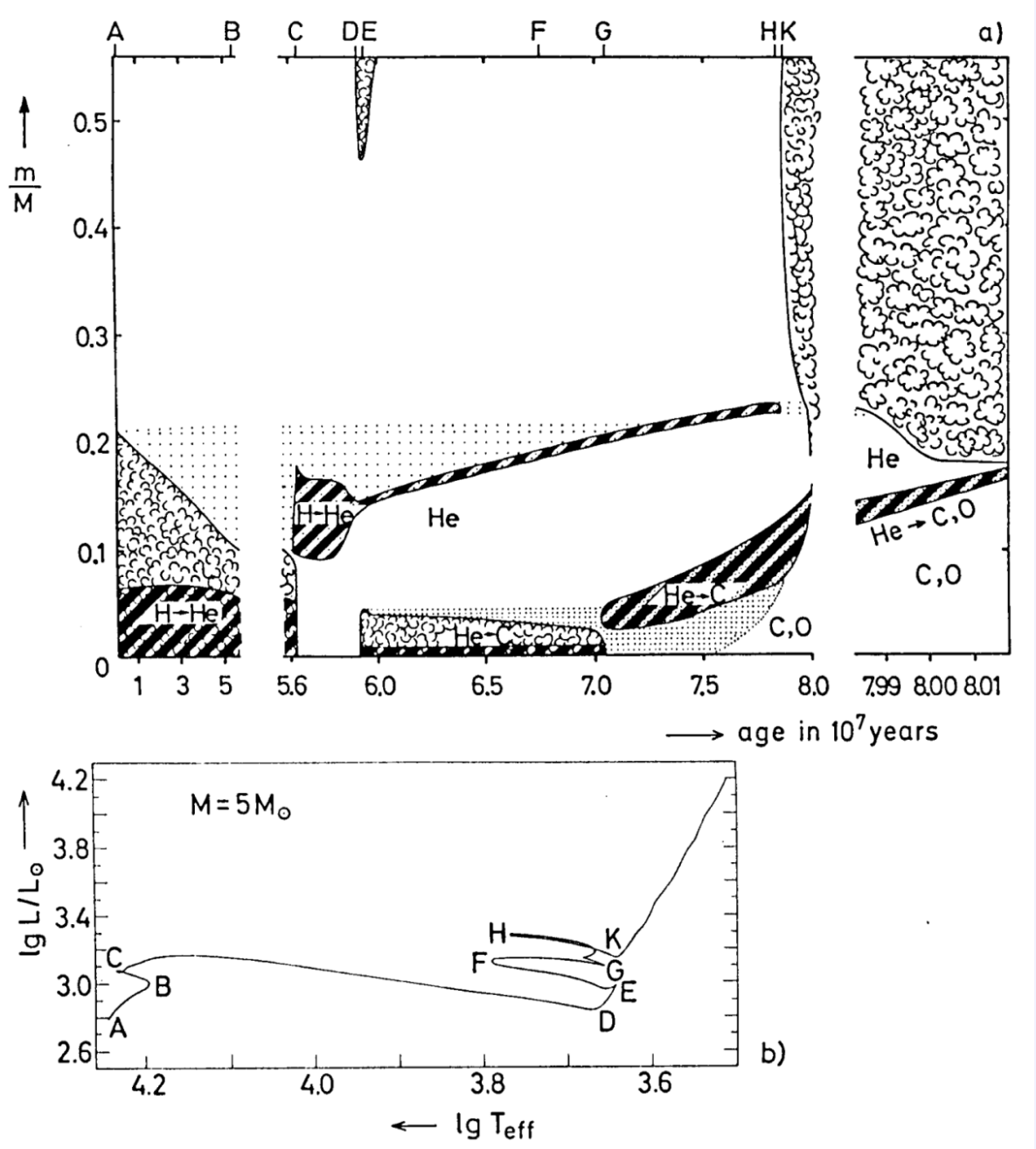
Outline

Moderate-mass star evolution
- Interior changes
- Core helium fusion
- Two shell sources
- Asymptotic Giant Branch
- Planetary nebulae and white dwarfs

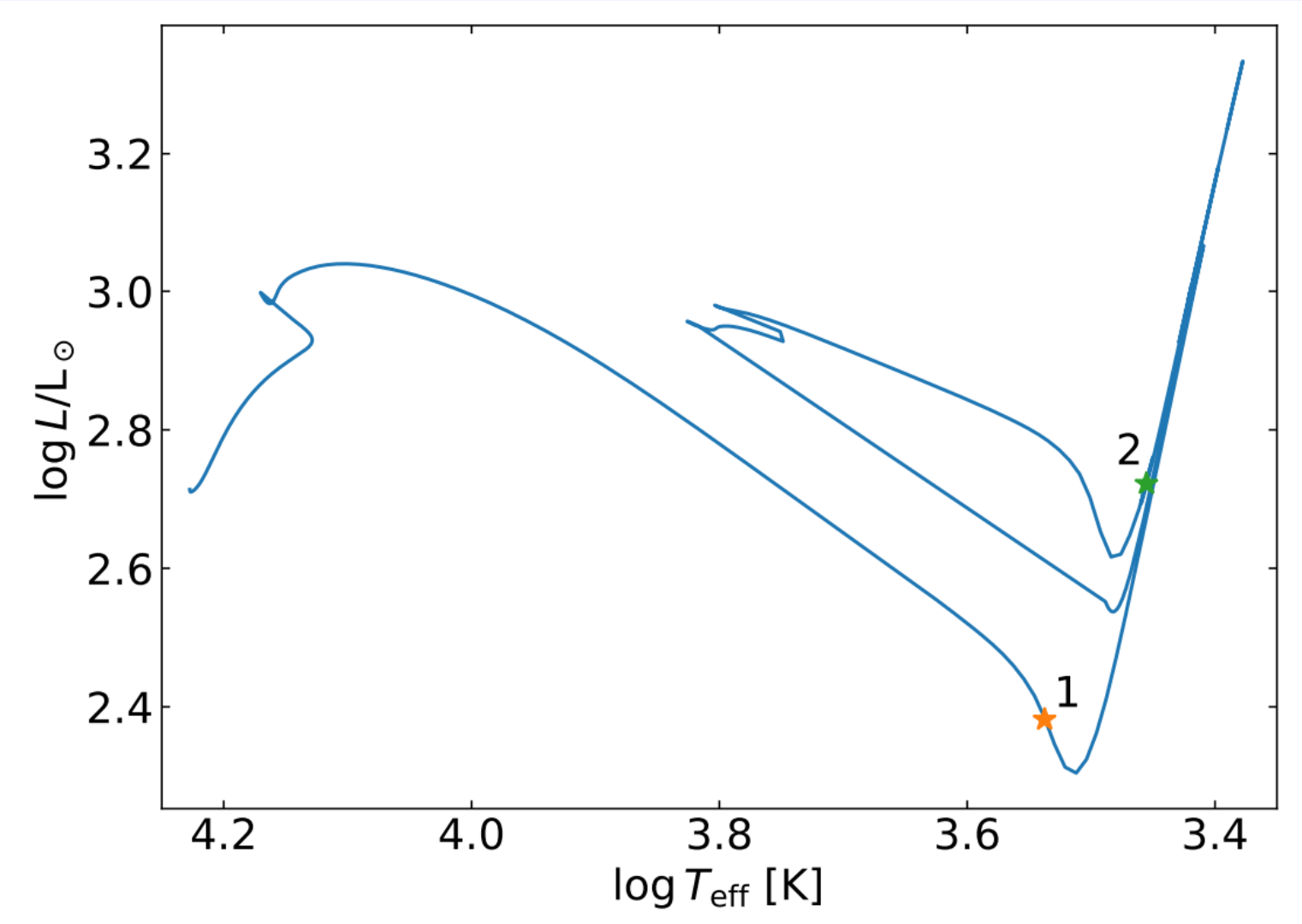
Summary

Moderate-mass star evolution - interior changes



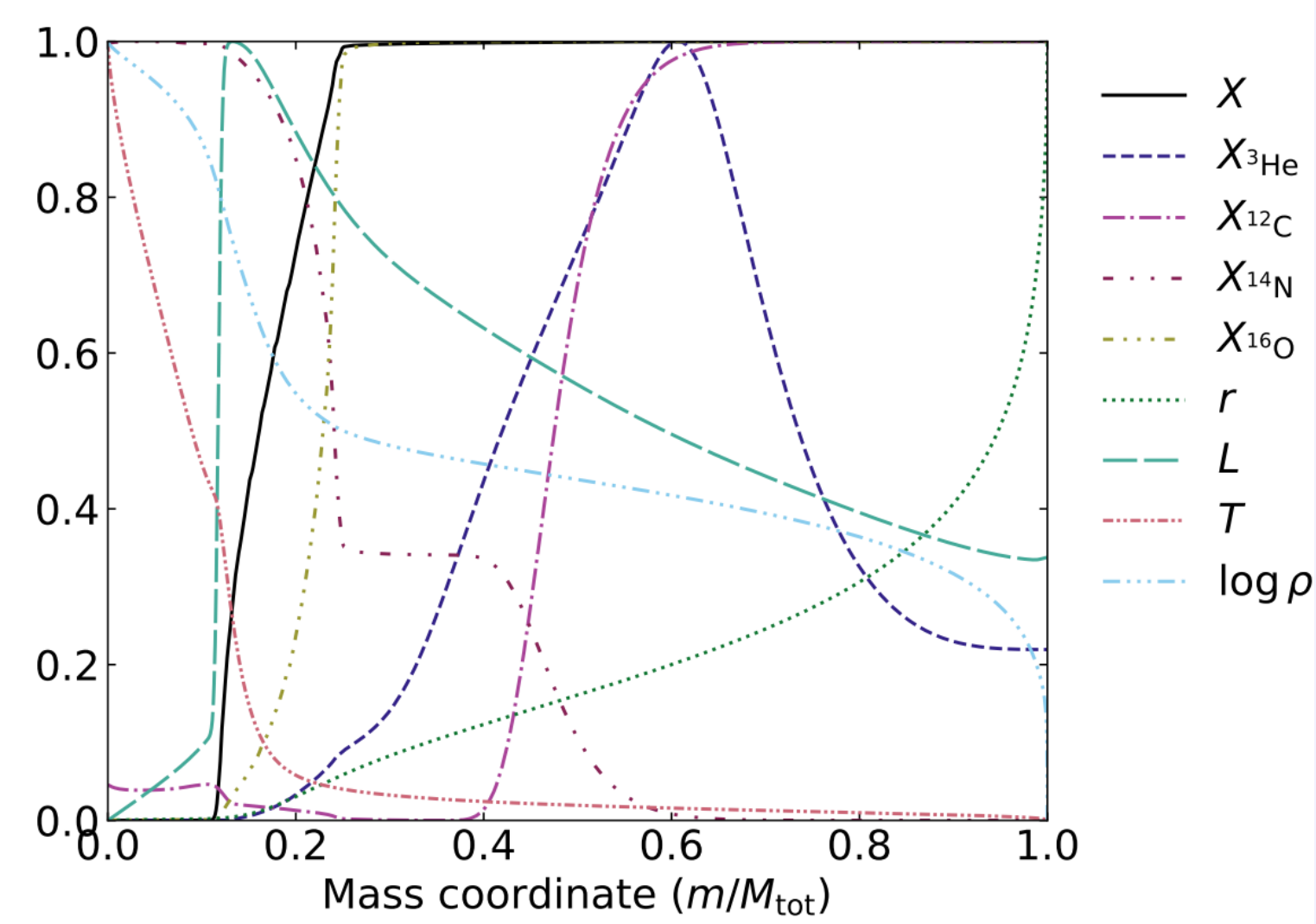
Credit: JCD - Fig 12.3

Moderate-mass star evolution - MESA model



Credit: JCD - Fig 12.4

Moderate-mass star evolution - property changes (1)



Credit: JCD - Fig 12.5 - check caption for scale

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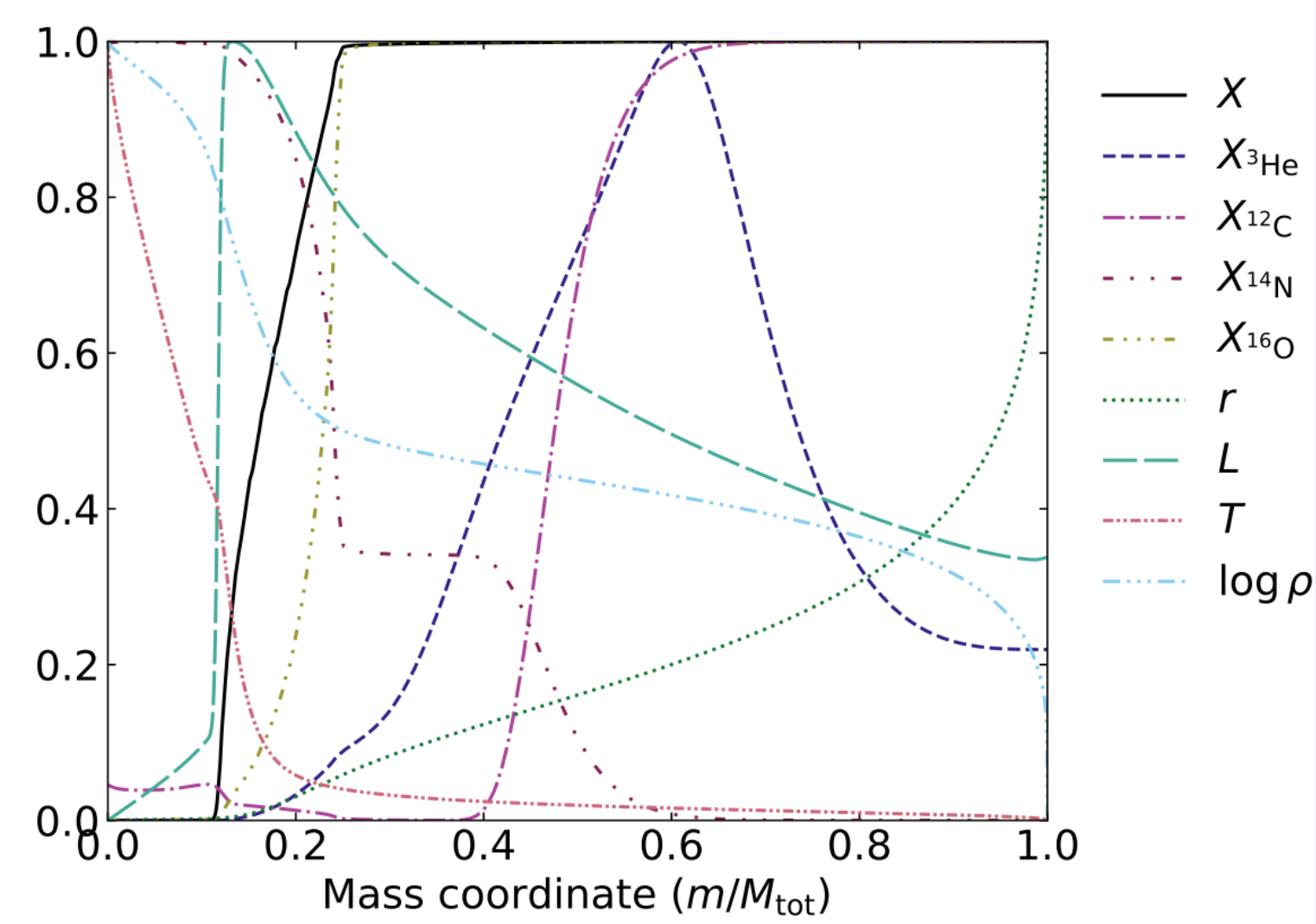
Moderate-mass star evolution - property changes (1)



Credit: JCD - Fig 12.5 - check caption for scale

Core He fusion

- The contracting helium core will generate energy
- This energy needs to be transported
- Hence a temperature gradient will need to come back
- Core is no longer isothermal and heats up
- If core temperature gets high enough, He fusion starts
- He fusion happens in a convective core

Repeating evolution

- Due to the start of core He fusion, the core will expand
- As the H shell is still present, this means that the envelope and thus star will contract
- The star moves back down the Hayashi track
- When the convective envelope shrinks, the effective temperature will rise again
- As the star is again dominated by radiative energy transport, the mass-luminosity relation holds again

Established core He burning

- When the He fusion is established, the core expansion and star contraction will stop
- Extended stable period of core He burning
- Note it is still much faster than the core H burning

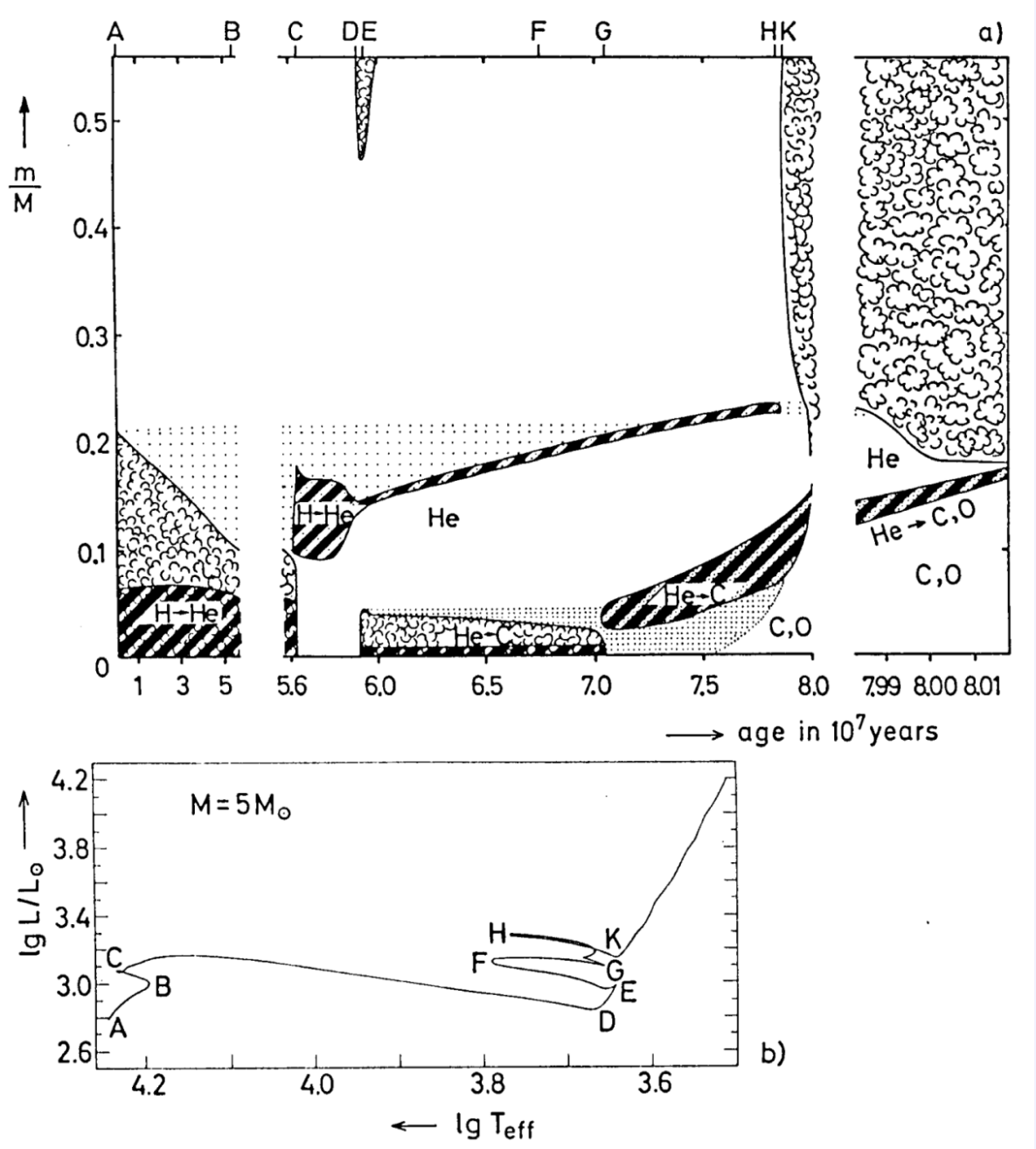
The end of core He fusion

- When He fusion is nearing the end, the core has increased its mean molecular weight
- It will start to contract again and thus heat up
- H shell is still in place, thus the star will expand again
- The star is back on its way to the Hayashi track

Two shell sources

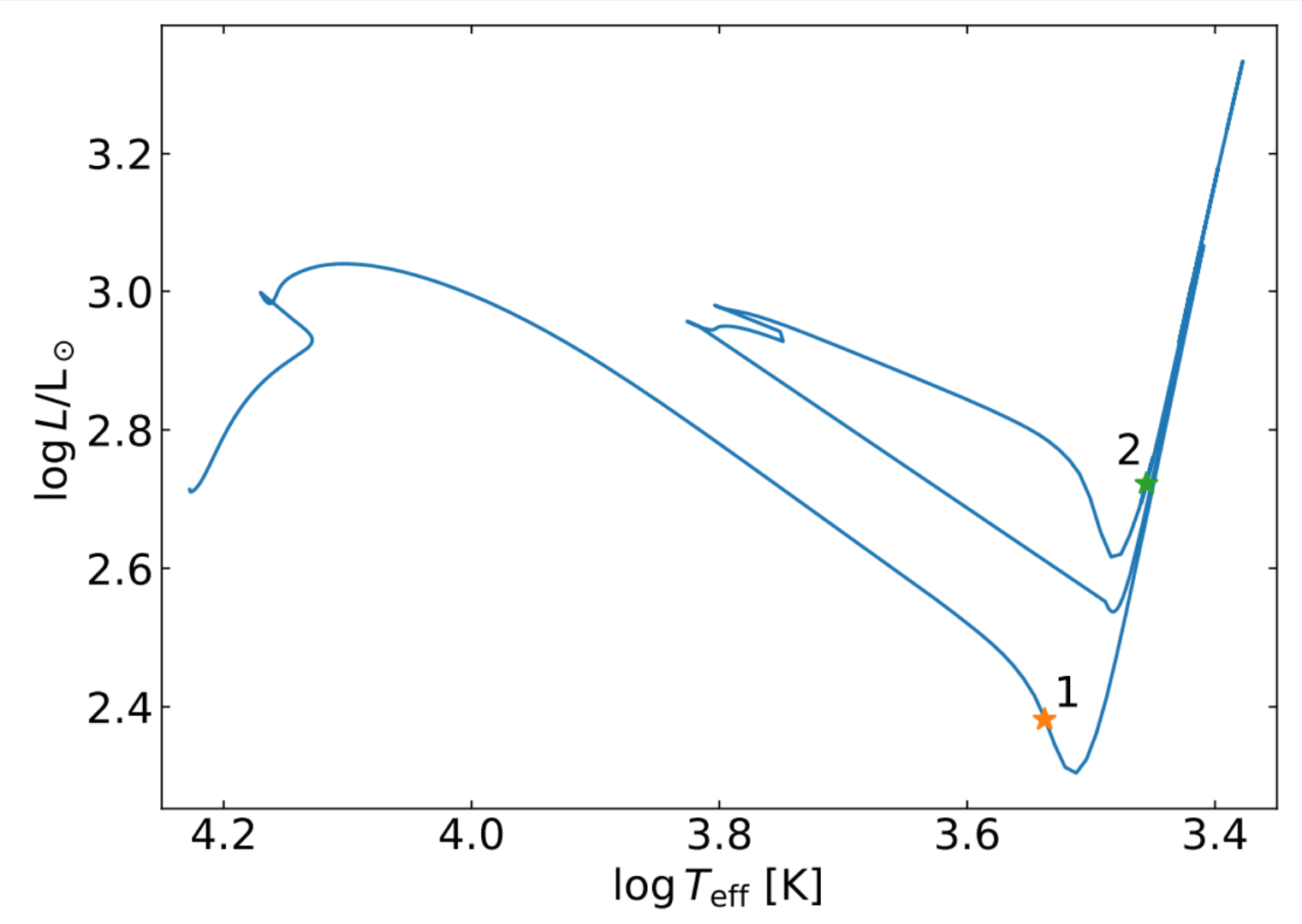
- When He is exhausted in the core, we now have a C-O core and 2 shell burning regions
- The C-O core is not yet hot enough and thus inert
- Core contracts, but now there are 2 shell sources!
- We could apply the shell burning law now, where the contracting core will make the star contract this time, but recent calculations paint a slightly different picture

Moderate-mass star evolution - interior changes



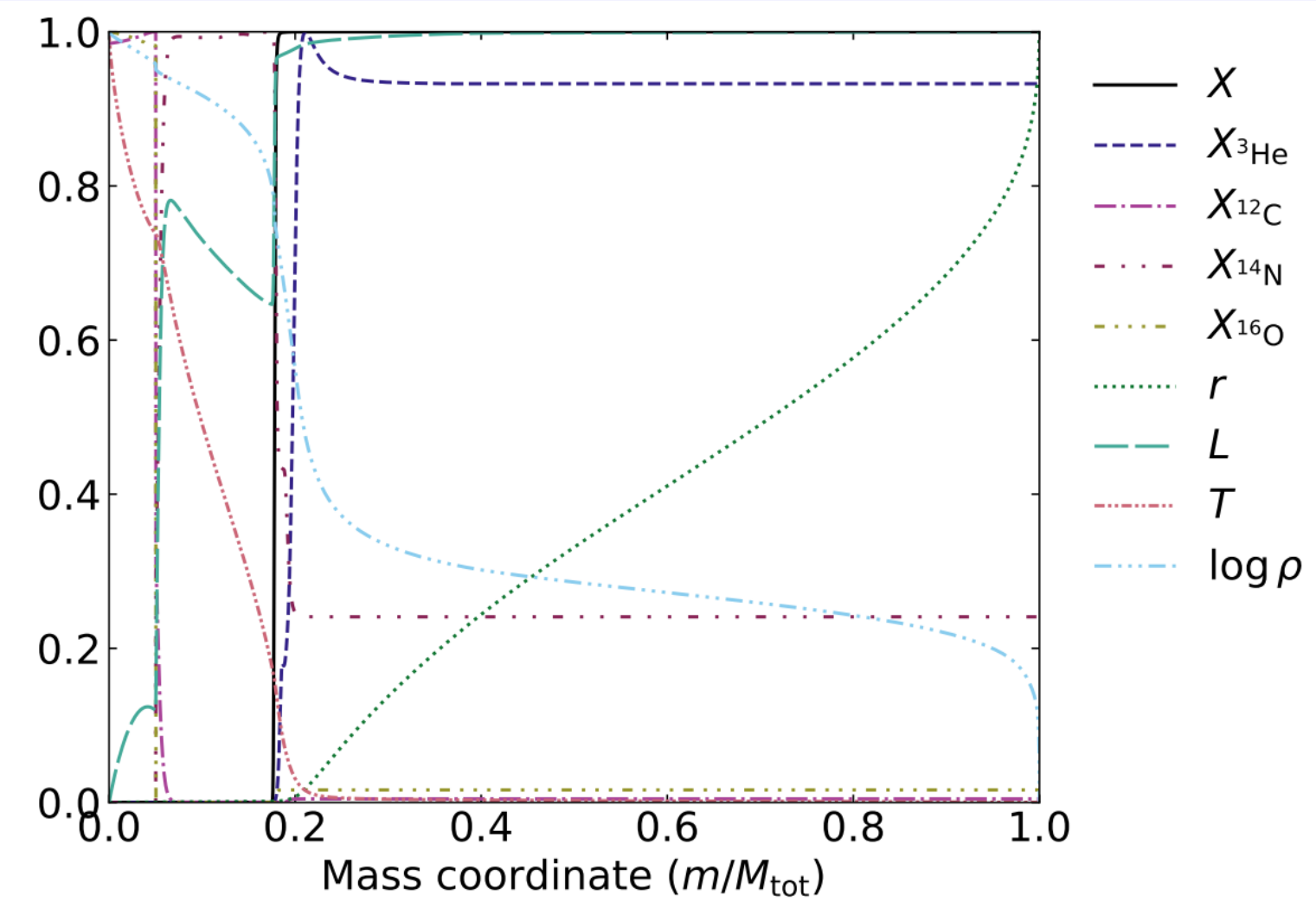
Credit: JCD - Fig 12.3

Moderate-mass star evolution - MESA model



Credit: JCD - Fig 12.4

Moderate-mass star evolution - property changes (2)



Credit: JCD - Fig 12.6 - check caption for scale

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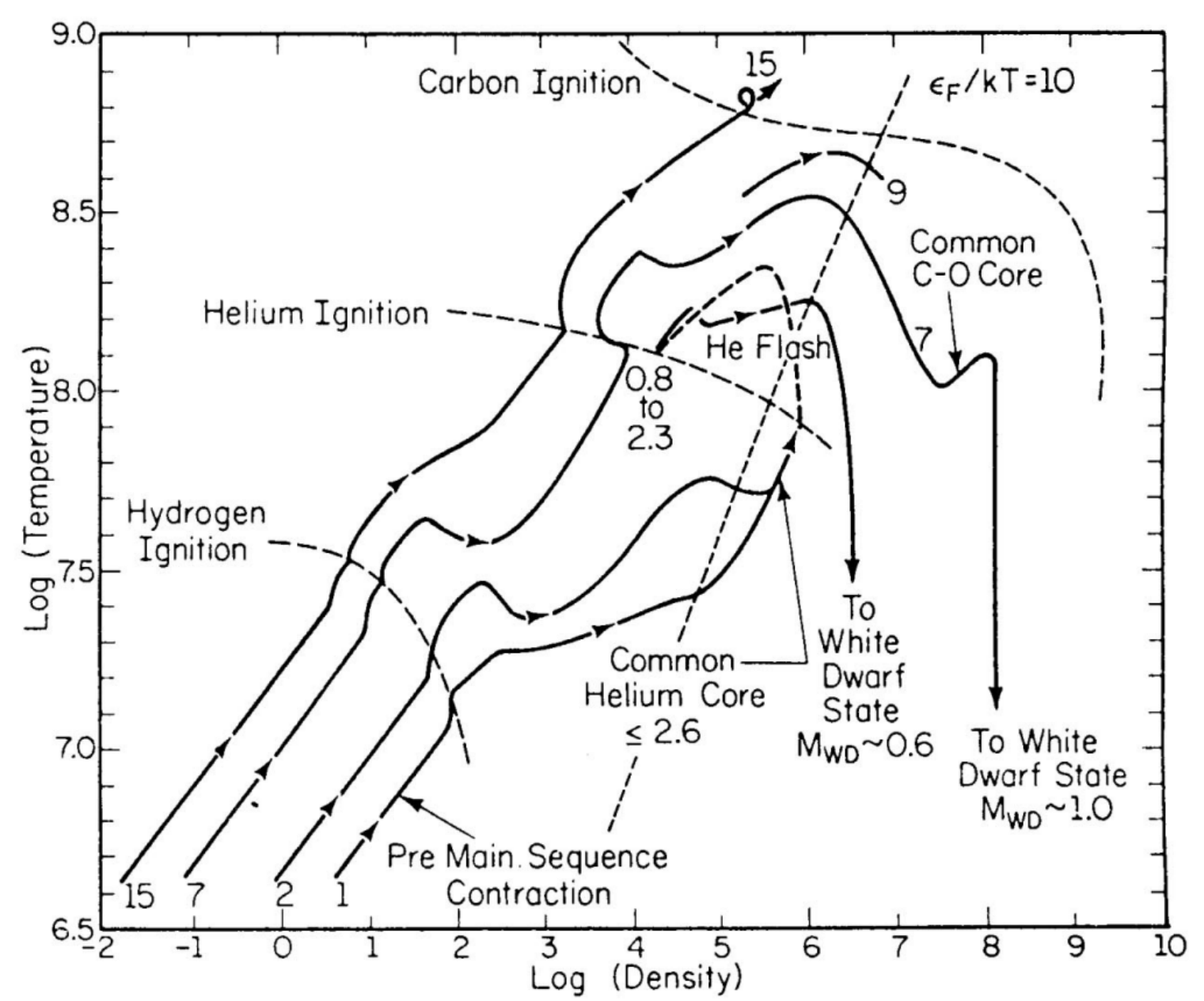
Degenerate cores

- For stars lower than about 10 solar masses, the C-O core will never reach conditions for C fusion to start
- The core reaches a state of electron degeneracy
  - Remember lecture 4 and see JCD3.5
  - Electrons become degenerate in high density environments when quantum-mechanical effects are needed to describe matter.
  - The Pauli exclusion principle (at most 2 fermions in given quantum state) means not enough low energy states available in dense environments.
  - This leads to pressure not being able to explained as ideal gas and now solely relating to density
- Due to the electron degeneracy, the temperature stops increasing
- This is the end of nuclear fusion in the cores, but the H and He shells are still burning

Stellar evolution - degenerate cores

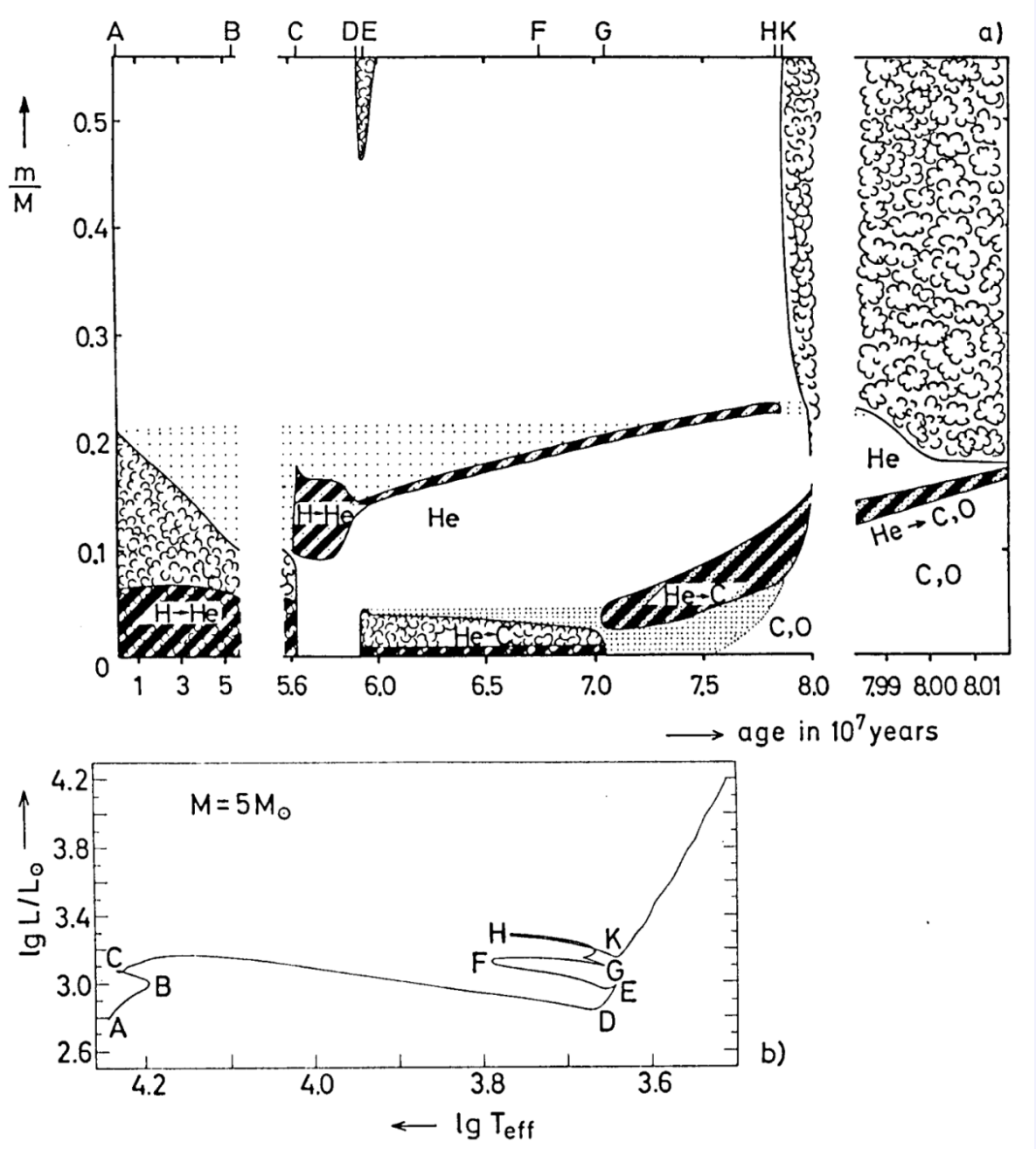


Credit: JCD - Fig 12.7

Asymptotic giant branch

- Once He shell burning is fully established, the region between the two shells keeps expanding
- This eventually extinguishes the H shell
- With only one shell source now and still the contracting core, the star expands again
- The temperature decreases till the Hayashi track and then the star moves up it again
- This phase is called the asymptotic giant branch
- During this phase a very deep convective envelope is established
- This can now mix - or dredge up - the new nuclear materials to the surface and change the surface composition of the star

Moderate-mass star evolution - interior changes



Credit: JCD - Fig 12.3

Subsequent evolution

- What happens now is highly complex and not well understood
- Through neutron capture, heavier elements are produced - these are observationally found in spectra of AGB stars
- Eventually, the star will start to loose mass, and in particular all the material outside the C-O core
- Due to this envelope contraction, the star rapidly moves left in the HR diagram
- End result is first a planetary nebula - hot and luminous core that illuminates the surrounding material
- And finally, when all the material is dispersed, a gradually cooling core, called a white dwarf

The Helix planetary nebula

Credit: ESA Hubble

The Ring planetary nebula

Credit: ESA/Webb, NASA, CSA, M. Barlow (University College London), N. Cox (ACRI-ST), R. Wesson (Cardiff University)

Summary

- Core He fusion happens in a convective region and somewhat repeats the process of H core burning
- If core He is exhausted, there is an inert C-O core and 2 shell burning regions
- C-O Core becomes degenerate for stars less massive than 10 solar masses
- Nuclear fusion ends and the star moves up the AGB
- Through not-well understood processes, the star loses its mass outside the C-O core, revealing a planetary nebula and finally a white dwarf

NEXT LECTURE: Solar mass stellar evolution, He flash